

Other Interpolation Formulae

For Lerp, in general, if we have p data points we can fit a polynomial of degree $p-1$ through these points:

$$y = \sum_{r=0}^{p-1} a_r x^r$$

We have p unknown coefficients and need p equations to find them

$$\hat{y}(x) = \frac{(x-x_1)y_2 + (x_2-x_1)y_1}{x_2-x_1} = y_1 L_1(x) + y_2 L_2(x)$$

$$L_1(x), L_2(x) = \frac{(x-x_2)}{(x_1-x_2)}, \frac{(x-x_1)}{(x_2-x_1)}$$

$$L_K(x) = \prod_{r=1}^p \frac{(x-x_r)}{(x_K-x_r)}, \quad r \neq K$$